

PHANPHUM PRATHUMSUWAN

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EDUCATION

Mahidol University, Faculty of Information and Communication Technology (ICT),
Bachelor of Science in Software Engineering

Aug 2022 – May 2026

EXPERIENCE

- Full-Stack Developer — DormiSpace: Dormitory Management PlatformAug 2025 – Apr 2026
- Designed and developed scalable RESTful APIs using Spring Boot for authentication, contract lifecycle management, and a multi-tenant billing system, with secure role-based access control and optimized PostgreSQL schemas.
 - Led Flutter frontend development for the mobile application, translating UI/UX requirements into maintainable screens, reusable components, and iterative feature improvements.
 - Diagnosed production-level issues involving API responses, data integrity, and system reliability, while maintaining GitHub Actions CI/CD pipelines to support stable release delivery.
- Research Intern at ReaLearn Laboratory — Tohoku University, JapanMay 2025 – Aug 2025
- Designed and implemented an automated benchmarking framework to evaluate reliability and failure behavior across seven LLMs, improving evaluation coverage and support for robust AI deployment.
 - Analyzed model outputs to identify failure patterns and reliability gaps, producing insights that informed more stable AI system behavior.
- Research Intern at Ishii Laboratory — Japan Advanced Institute of Science and Technology (JAIST), JapanJun 2025 – Aug 2024
- Applied formal verification techniques using Lustre/Kind2 and Simulink/SLDV to analyze system correctness, safety properties, and complex system behavior.
 - Translated and modelled system specifications across platforms, identifying state-space complexity limitations and producing insights to support technical documentation, system planning, and design decisions.
- Participant in Mahidol AI Hackathon 2 — Faculty of Information and Communication TechnologyMar 2024
- Collaborated in a cross-functional team to design and implement an AI-based solution for addressing healthcare service challenges
 - Managed data preparation and model integration, ensuring smooth interaction between data pipelines and application logic

PROJECTS

- GoodPoliz: AI-Assisted Police Operations System
- Implemented Flutter automated UI tests using the integration_test framework and Flutter Drive runner on Flutter Web (Chrome), covering happy and unhappy path scenarios across three core workflows
 - Wrote backend unit tests (JUnit + Mockito) applying input space partitioning and logic coverage techniques; conducted manual system testing and produced a requirement traceability matrix to ensure test coverage aligned with functional requirements
- Gemini Telescope Control System
- Designed and implemented a web-based control system integrating modern RESTful APIs with a legacy Observatory Control System (OCS), enabling seamless communication between new and existing components
 - Modelled system architecture using UML diagrams to support technical design planning and stakeholder
- Online Tutor Search and Management Website
- Designed and implemented backend logic and relational database schemas (SQL) for managing tutor profiles and booking workflows
 - Optimized query performance and matching algorithms to improve system efficiency and response time

SKILLS & LANGUAGES

Programming: C, Java, Python, Dart, JavaScript, SQL
Backend & Database: Spring Boot, RESTful API Design, PostgreSQL, MySQL, Firebase
Cloud & Third-Party Services: AWS, Google Cloud Vision OCR, Omise Payment Gateway
Testing & DevOps: JUnit, Mockito, Flutter Unit/Integration Testing, Robot Framework, JMeter, GitHub Actions, Docker
Systems & Reliability: Linux, Kibana, application monitoring, incident troubleshooting, root cause analysis
Languages: Thai (Native), English (IELTS 7.0, TOEIC 895), Japanese (JLPT N5)